OPERATING AND FINANCIAL LEVERAGE

Practical Questions with Answers

Source: SBPD Publications – Financial Management

Section	Questions
A. Operating Leverage	Q.1 – Q.11
B. Financial Leverage	Q.12 – Q.23
C. Combined Leverage	Q.28 – Q.33

Note: *Amounts are in (Indian Rupees). Answers shown are those given in the textbook. Questions 24–27 are not included in the pages provided; Q.23, Q.28 and Q.33 are partly cut off in the scans and are marked accordingly.*

A. Operating Leverage

Q.1

A company has estimated that for a new product its selling price is 15 per unit, variable cost is 10 per unit and fixed cost is 10,000. Calculate the operating leverage for sales volume of 4,000 units and 5,000 units.

Answer: 2; 1.67

Q.2

X Ltd. sells its products for 500 per unit. Its variable cost is 200 per unit and fixed costs are 1,00,000 per year. Show the various levels of EBIT that would result from sale of (i) 2,000 units, (ii) 2,500 units, (iii) 1,500 units. Calculate also operating leverage.

Answer: Case I: 1.20; Case II: 1.29; Case III: 1.15

Q.3

A company has estimated that for a new product its selling price is 56 per unit, variable cost is 36 and fixed cost is 40,000. Calculate the operating leverage for sales volume of 12,000 units and 16,000 units.

Answer: 12,000 units: 1.2; 16,000 units: 1.14

Q.4

The installed capacity of a factory is 8,000 units. Actual capacity used is 4,000 units. Selling price per unit is 50 and variable cost is 30 per unit. Calculate the operating leverage in each of the following three situations:

(i) When fixed costs are 4,000.

(ii) When fixed costs are 10,000.

(iii) When fixed costs are 12,000.

Answer: I: 1.05; II: 1.14; III: 1.18

Q.5

The annual sales of a company are 3,00,000 units. The selling price is 10 per unit, variable cost 6 per unit and fixed costs are 4,00,000. In case the sales increase by 10%, then what will be the increase in earnings before interest and tax (EBIT) as a result of operating leverage?

Answer: 15%

Q.6

The following estimates are given for a new product:

Particulars	Estimate
Break-even point	2,000 units
Variable cost	9 per unit
Selling price	14 per unit

Calculate the operating leverage from the above estimates for a volume of 2,500 units and 3,000 units and draw suitable inference from the difference in operating leverages for the two output levels.

Answer: OL = Nil, 5, 3

Q.7

The following estimates are given for a new product:

Particulars	Estimate
Break-even point	20,000 units
Variable cost	50 per unit
Selling price	70 per unit

Calculate operating leverage from the above estimates for a volume of 25,000 units and 30,000 units and draw suitable inference from the difference in operating leverage for the two output levels.

Answer: OL = 5 and 3

Q.8

(a) If operating leverage is 2 and sales increase by 50%, then by what percentage of EBIT will increase?

(b) If operating leverage is 3 and the firm wants to double its EBIT, how much rise in sales would be needed on a percentage basis?

Answer: (a) 100%; (b) 33.33%

Q.9

A firm sells its products for 100 per unit. Its variable cost is 50 per unit and fixed costs are 25,000 per year. Show the various levels of EBIT that would result from sale of (i) 1,000 units, (ii) 1,250 units and (iii) 750 units. Calculate also operating leverage.

Answer: OL: Case I = 2; Case II = 3; Case III = 1.67

Q.10

There are three companies 'A', 'B' and 'C'. The volume of production in each company is 50,000 units. Price per unit is 10 and variable costs per unit are 4 for all the three companies. Fixed costs are 1,00,000 in company 'A', 2,00,000 in company 'B' and 2,50,000 in company 'C'. It is assumed that the entire output is sold in the market. You are required to find out the operating leverage in each of the companies. If there is 10% increase or 10% decrease in the sales income of each company, find out the impact of such increase or decrease on the operating profits (OP) or earnings before interest & tax (EBIT) of each company.

Answer: Operating leverage: A = 150%, B = 300%, C = 600%. Impact on operating profit: Company A = +15%, –15%; Company B = +30%, –30%; Company C = +60%, –60%

Q.11

You are given the following information:

Particulars	Amount
Selling Price per Unit (P)	10
Variable Cost per Unit (VC)	6
Fixed Costs (FC)	16,000

Calculate break-even point and demonstrate how DOL varies with the volume of output above or below the BEP level.

Answer: (i) 4,000 units; (ii) 5 times; (iii) 3 times; (iv) 2 times; (v) (–) 3 times

B. Financial Leverage

Q.12

Priyank Ltd. Company has a choice of the following three financial plans. You are required to calculate the financial leverage in each case:

Particulars	A ( )	B ( )	C ( )
Equity Capital	1,00,000	50,000	1,50,000
Debt @ 10%	1,00,000	1,50,000	50,000
Operating Profit (EBIT)	20,000	20,000	20,000

Answer: A = 2; B = 4; C = 1.33

Q.13

Supreme Ltd. has the following capital structure:

Particulars	Amount ( )
Equity Share Capital	4,00,000
10% Preference Share Capital	4,00,000
8% Debentures	10,00,000

The present EBIT is 2,00,000. Calculate the financial leverage assuming that the company is in 50% tax bracket. What would be the new financial leverage if operating profit increases to 3,00,000?

Answer: 5; 2.14

Q.14

Siyaram Ltd. has the following capital structure:

Particulars	Amount ( )
50,000 Equity Shares of 10 each	5,00,000
10,000, 10% Pref. Shares of 100 each	10,00,000
10,000, 10% Debentures of 100 each	10,00,000

Calculate the EPS for each of the following levels of EBIT: (i) 5,00,000, (ii) 3,00,000, (iii) 7,00,000. The rate of income-tax for the company is 50%. In the above case, also calculate the financial leverage taking EBIT 5,00,000 as base.

Answer: EPS = 2, Nil, 4; FL: Case I = 2.5, Case II = 2.5

Q.15

A company has sales of 10,00,000, variable costs 7,00,000, fixed costs 2,00,000 and debentures 5,00,000 at 10% interest. Calculate: (i) Operating leverage, (ii) Combined leverage.

Answer: (i) OL = 3; (ii) CL = 6

Q.16

The financial manager of Y Ltd. expects that its earnings before interest and taxes in the current year would be 30,000. The company has issued 5% debentures of 1,20,000, while 10% preference shares amount to 60,000. It has 3,000 equity shares of 10 each. What would be the degree of financial leverage assuming the EBIT (i) 18,000 and (ii) 42,000. The tax rate of the company may be taken as 50%. How would EPS be affected?

Answer: Case I = 2.5 (Unfavourable); Case II = 2.5 (Favourable)

Q.17

A company has the following capital structure:

Particulars	Amount ( )
Equity Share Capital	1,00,000
10% Preference Share Capital	1,00,000
8% Debentures	1,25,000

The present EBIT is 50,000. Calculate the financial leverage assuming that the company is in 50% tax bracket.

Answer: FL = 2.5

Q.18

The financial manager of a company expects that its earnings before interest and tax (EBIT) in the current year would amount to 50,000. The company has issued 5% bonds aggregating 2,00,000 and the 10%

preference shares amounted to 1,00,000. The company has 5,000 equity shares of 10 each. What would be the degree of Financial Leverage (DFL)? Assuming the EBIT (a) 30,000 and (b) 70,000. The tax rate of the company may be assumed at 50%. How would EPS be affected?

Answer: Case I = 2.5 (Unfavourable); Case II = 2.5 (Favourable)

Q.19

(a) If financial leverage is 1.5 and EBIT increases by 10%, then what percentage of EBT will increase?

(b) If financial leverage is 2 and EPS increases by 10%, then what percentage of EBIT will increase?

Answer: (a) 15%; (b) 5%

Q.20

The following data are available in respect of a company:

Particulars	Amount ( )
Equity Share Capital (divided into shares of 10 each)	50,00,000
Debentures (12%)	20,00,000
Earnings before Interest & Tax (EBIT)	16,00,000

You are required to calculate the Financial Leverage (FL) of the company.

Answer: FL = 1.176

Q.21

Find out financial leverage from the following data:

Particulars	Value
Net Worth	25,00,000
Debt/Equity	3 : 1
Interest Rate	12%
Operating Profit	20,00,000

Answer: FL = 1.818

Q.22

A company has the following capital structure:

Particulars	Amount ( )
2,00,000 Equity Shares of 10 each	20,00,000
40,000 10% Pref. Shares of 100 each	40,00,000
40,000 10% Debentures of 100 each	40,00,000

Calculate the EPS for each of the following levels of EBIT: (i) 20,00,000; (ii) 12,00,000; (iii) 28,00,000. The rate of income tax for the company is 50%. In the above case, also calculate the financial leverage taking EBIT 20,00,000 as base.

Answer: EPS: (i) 2.00, (ii) Nil, (iii) 4.00; FL: (i) 2.5, (ii) 2.5

Q.23

Below is given the Balance Sheet of Parijat Company Ltd.:

Liabilities	₹	Assets	₹
Equity Share Capital (shares of 10 each)	1,80,000	Fixed Assets	4,50,000
Surplus & Reserves	60,000	Current Assets	1,50,000

(The balance sheet and the rest of this question are cut off in the scanned page.)

C. Combined Leverage

Q.28

(Question text is only partly visible in the scan.) On the basis of the above data, compute: (i) Degree of Operating Leverage, (ii) Degree of Financial Leverage, (iii) Degree of Combined Leverage.

Answer: DOL = 2.5; DFL = 1.6; DCL = 4

Q.29

Rangnath Udyog Limited has annual sales of 2,50,000, variable costs 1,70,000 and fixed costs 40,000. The firm uses a debt of 1,00,000 on which interest @ 12% per annum is payable. On the basis of the above data compute (i) Operating Leverage, (ii) Financial Leverage and (iii) Combined Leverage.

Answer: (i) 2; (ii) 1.43; (iii) 2.86

Q.30

The following projections are related to companies A and B:

Particulars	A	B
Output and Sales (Units)	1,60,000	2,00,000
Variable Cost per Unit ( )	4	3
Fixed Costs ( )	4,80,000	5,00,000
Interest Burden on Debt ( )	2,40,000	1,00,000
Selling Price per Unit ( )	10	8

On the basis of the above data, compute (i) Operating leverage, (ii) Financial leverage, (iii) Combined leverage.

Answer: OL: A = 2, B = 2; FL: A = 2, B = 1.25; CL: A = 4, B = 2.50

Q.31

The following projections are related to companies Disha and Asha:

Particulars	Disha Co.	Asha Co.
Output & Sales (Units)	45,000	55,000
Variable Cost (per unit) ( )	4	3

Fixed Cost ( )	1,25,000	1,30,000
Interest on Debts ( )	62,500	30,000
Selling Price (per unit) ( )	10	8

On the basis of the above data compute (i) Operating leverage, (ii) Financial leverage, (iii) Combined leverage.

Answer: (i) OL = 1.862, 1.897; (ii) FL = 1.758, 1.261; (iii) CL = 3.273, 2.391

Q.32

The total annual sales of a company are 5,00,000. Its variable costs are 40% of sales and fixed costs are 2,00,000. The annual interest burden on borrowed capital is 60,000. You are required to calculate the operating leverage, the financial leverage and the combined leverage of the company. What will be the percentage increase in profits before tax in case the company's sales increase by 10 per cent? Verify by taking an example.

Answer: OL = 3; FL = 2.5; CL = 7.5; % increase in profits = 75%

Q.33

The following estimates are available in respect of a company:

Particulars	Amount ( )
Sales	10,00,000
Less: Variable Costs	4,00,000
Contribution (C)	6,00,000
Less: Fixed Costs	4,00,000
Earnings Before Interest & Tax (EBIT)	2,00,000
Less: Interest on Debt Capital	1,00,000
Profits Before Tax (PBT)	1,00,000

(The question's requirements continue on the next page, which is not included in the scans.)